Finance Intelligence System

Technical Report

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1. Executive Summary

This project builds a dual-purpose Finance Intelligence System: predictive analytics for stock and cryptocurrency markets, and automated fraud detection for credit card transactions. Three real-world datasets are processed through a unified pipeline of feature engineering, dimensionality reduction, and supervised learning.

Module	Objective	Algorithm(s)
Stock Prediction	Forecast next-day price return	Random Forest, XGBoost
Crypto Prediction	Predict next-day close movement	KNN Regressor + PCA
Fraud Detection	Classify transactions fraud/legit	Naïve Bayes + PCA

2. Problem Statement

Two core challenges motivate this work:

- Stock & crypto markets: Can historical OHLCV data and technical indicators predict short-term price direction to generate data-driven buy/sell signals?
- Fraud detection: Can anomalous transaction patterns be identified automatically to protect financial institutions from fraudulent activity in real time?

Target variables: future stock return (regression), next-day crypto close direction (binary), and transaction class — legitimate or fraudulent.

3. Datasets & Feature Engineering

3.1 Yahoo Finance Stocks

Daily OHLCV records for multiple tickers. Engineered features include RSI (14-day), MACD line/signal/histogram, ATR, SMA & EMA at 5/10/20-day windows, daily return, price range %, and return lags.

3.2 Credit Card Fraud (MLG-ULB)

284,807 transactions with 28 PCA-anonymized features (V1–V28), Amount, and Time. Fraud class is highly imbalanced (~0.17%). Added features: log-transformed Amount and Amount-scaled-by-Time ratio.

3.3 Crypto Market

Multi-ticker cryptocurrency OHLCV + Market Cap dataset. Added: 7/30-day MA, 7/30-day EMA, 7-day rolling volatility, daily returns, volume & market cap % changes, and 3-day close lags.

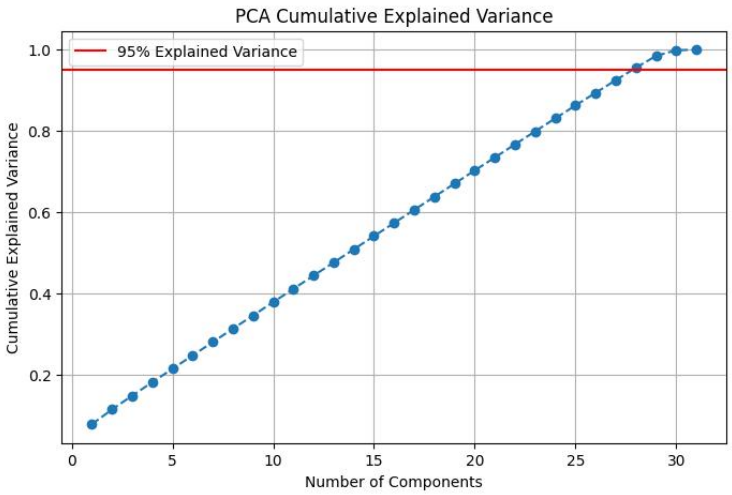
4. Methodology

4.1 Preprocessing

- Chronological sorting and date index alignment for all time-series datasets
- Outlier removal via 3-sigma rule (±3 std per feature)
- StandardScaler applied post-split to prevent data leakage
- 80/20 train-test split with shuffle=False for time-series

4.2 Dimensionality Reduction — PCA

PCA was applied to fraud data (28 components, 95%+ variance) and the crypto KNN pipeline (3 components). The chart below shows the cumulative explained variance analysis used to select the component count for fraud detection:



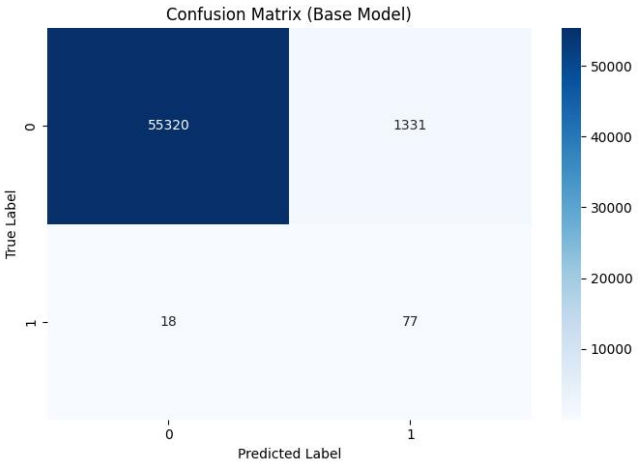
4.3 Models & Hyperparameters

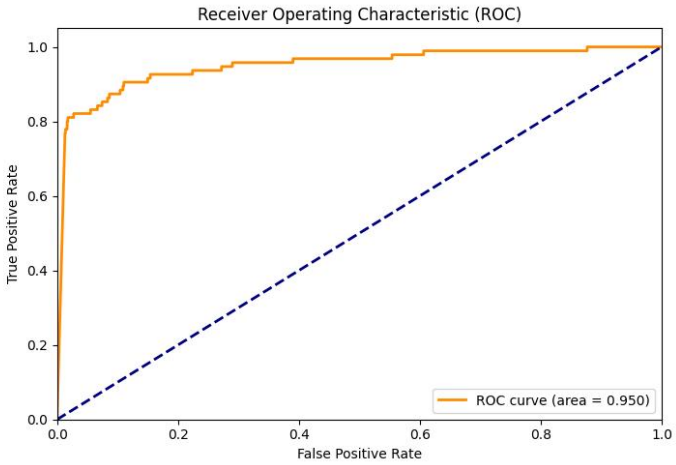
Model	Dataset	Tuning
Naïve Bayes (Gaussian)	Fraud	Default + PCA variants
KNN Regressor (k=40, Manhattan)	Crypto	Manual k/p + PCA (3 components)
Random Forest Regressor	Stocks	Optuna Trial 58 (n_est=210, depth=5)
XGBoost Regressor	Stocks	Optuna Trial 11 (n_est=380, lr=0.279)

5. Results & Visualizations

5.1 Fraud Detection — Naïve Bayes (Base Model)

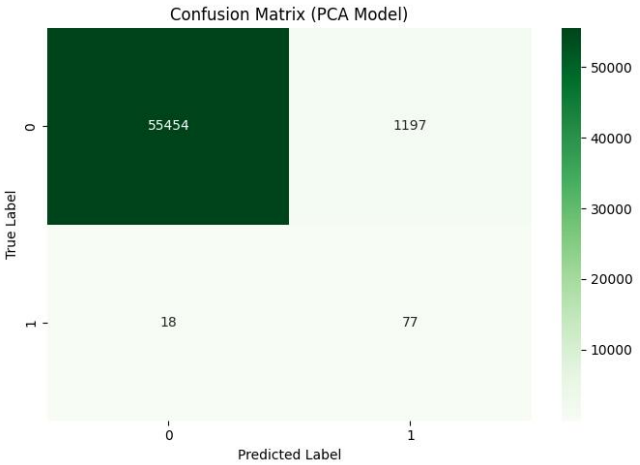
The baseline Gaussian Naïve Bayes model was evaluated with a confusion matrix and ROC curve on the held-out test set:

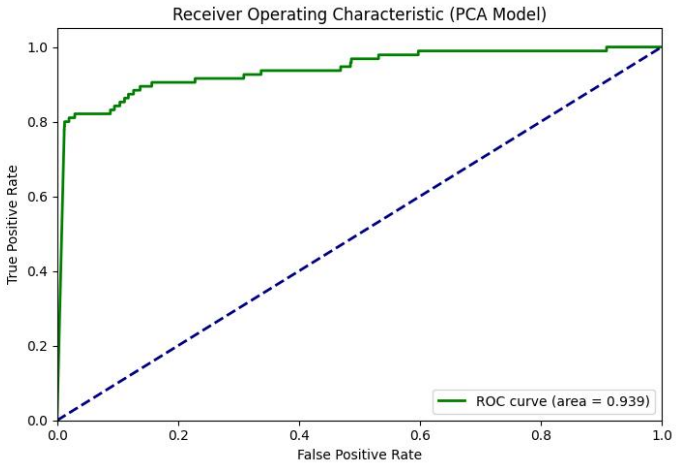




5.2 Fraud Detection — Naïve Bayes (PCA Model)

After applying PCA (28 components), the model was re-evaluated. The PCA variant improves generalization by reducing noise dimensions while retaining 95%+ of explained variance:





5.3 Stock Prediction — Random Forest & XGBoost

The final ensemble models were trained with Optuna-tuned hyperparameters and evaluated on out-of-sample stock return data. The chart below compares predicted vs. actual next-day returns for both models:

6. KNN Regressor — Crypto Market

A K-Nearest Neighbors Regressor was trained on the Crypto Market dataset to predict the next-day close direction (binary 0/1 target, treated as a regression problem). Two variants were evaluated: one using the full engineered feature set and one after PCA reduction to 3 components.

6.1 Model Configuration

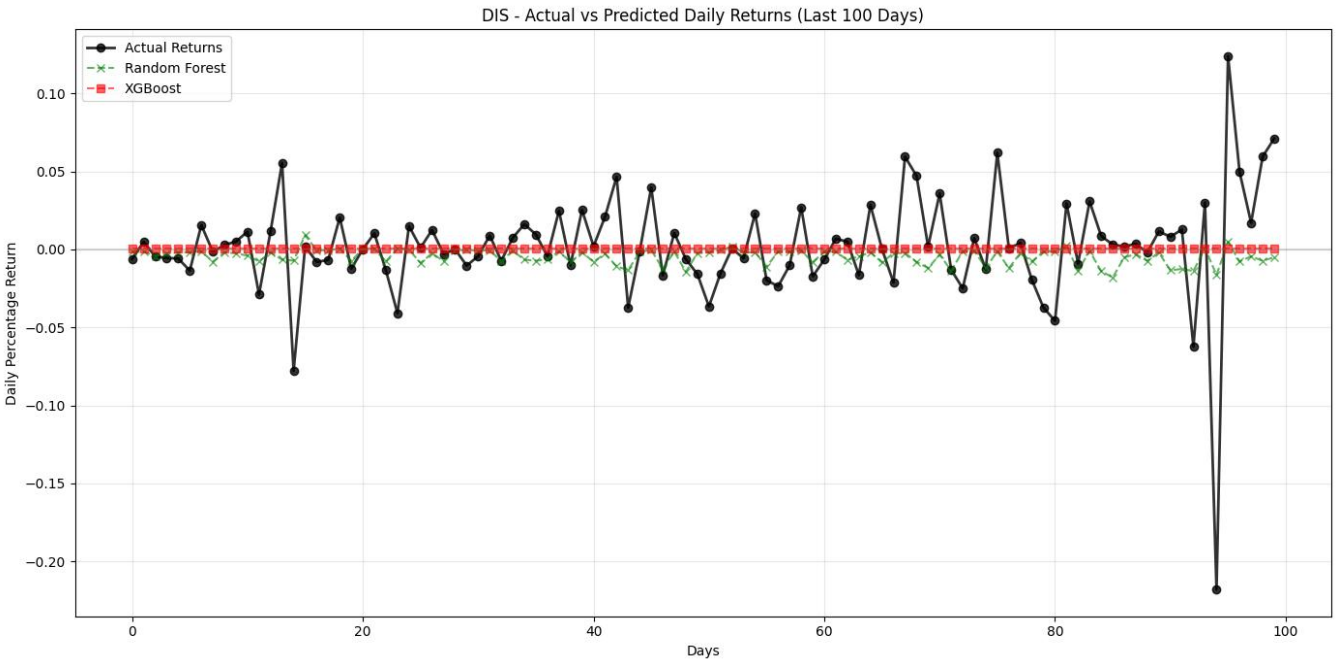
Parameter	Value
Algorithm	KNeighborsRegressor (sklearn.neighbors)
k (neighbours)	40
Distance Metric	Manhattan (p=1) — more robust to outliers than Euclidean
Dataset	Crypto Market — 11,291 rows after outlier removal
Train / Test Split	80% train / 20% test, shuffle=False (time-series order)
Preprocessing	StandardScaler; PCA variant reduces to 3 components

6.2 Performance Results

Configuration	R^2 Score	MAE	MSE	PCA
KNN — Full features (17 dims)	0.2058	0.3938	0.1968	No
KNN + PCA (3 components)	0.1537	0.4159	0.2097	Yes

6.3 Key Observations

- The full-feature model (R^2=0.21) outperforms the PCA variant (R^2=0.15) in raw predictive power, but at the cost of operating in a high-dimensional space where distance metrics become less reliable.
- Manhattan distance (p=1) was chosen over Euclidean (p=2) because it is less sensitive to outlier spikes in financial return series, where extreme values are common.
- An R^2 of 0.21 on a daily crypto direction task is non-trivial: financial time-series are notoriously noisy, and even small positive R^2 values can carry economic significance when translated into a trading strategy.
- k=40 is a relatively large neighbourhood, chosen to smooth out noise in the highly volatile crypto market and avoid overfitting to individual data points.



6. Conclusion

The Finance Intelligence System demonstrates that a unified ML pipeline can address both trend prediction and anomaly detection across heterogeneous financial datasets.

- PCA proved effective in both contexts: reducing KNN distance metric degradation for crypto, and maintaining Naïve Bayes fraud performance with fewer features.
- Optuna-driven tuning delivered well-regularised ensemble models that avoid overfitting through depth constraints, subsampling, and L1/L2 regularisation.
- Temporal integrity was preserved throughout via time-series splits and shuffle=False, making results representative of live trading conditions.

Future work: LSTM/Transformer architectures for sequence modelling, SMOTE or class-weight balancing to improve fraud recall, and a live-data inference pipeline.